Impact of Microlending Interest Rates

CS109 • Gerardo Murga Montelongo

Project App Link

Why I built this project:

I am an international student from Mexico. I live in a very large and tight knit family that allows for greater advantages, including being able to get small informal loans for 0% interest among members of the family if we don't want to get a traditional credit from a bank. This is a privileged situation because many **others who** don't have this familiar network and **cannot get access to traditional credit**, have to look out for **informal money lenders or microlending companies** who **often charge them high interest rates**.

Many people and small businesses in developing economies around the world have relied on these informal networks. At least that was until microlending as a way for institutions to provide loans to financially vulnerable populations rose as a concept in the 70's with the foundation of Grameen Bank. Whether microlending companies were doing good or wrong has remained a long-standing debate. Some companies, as they need to maintain their profitability, have begun charging super high interest rates arguing that the customers had faced similar interest rates through informal lending before the microlending companies even existed.

Through this interactive educational toy model, I am attempting to **portray the impact of high or low interest rates on financially vulnerable populations subject to microloans**, so that **students and citizens can accessibly understand** what microlending is and **how damaging high interest rates are** to these populations.

This is important to communicate because microlending and its consequences remain poorly understood, particularly in the developed countries that often build and export these financial models. We need to raise awareness that what these companies are doing causes real harm, and fight for international accountability; starting from developed economies like the US, so they can set an example for stronger protections in the developing economies most affected.

For my full argument focused on LATAM of why this is a problem, click here. (PWR2KTA Paper)

Description of the model:

I am simulating the alternative credit scoring system that many microlending companies perform as an alternative to more traditional FICO scores. The main reason behind this is that they are informal businesses with less accurate information. Model estimation of a user's credit score:

User's Credit score = *33% friends network + 33% payment history + 33% amount owed + total savings bonus.*

Many RVs used in the model are Gaussian distributions because by the **CLT**, quantities like # of texts with clients and # of businesses include many small independent behaviors and outcomes.

Friends Network estimates the network credit score by weighting each friend's credit score by their conditional probability of friendship and normalizing the result. The conditional probability of friendship P(Friends | Variables) is the most relevant part for this class. For each friend, P(Friends | Variables) is modeled as a weighted **additive probability based on a Bayes Net** that considers the prior. This branch of study is called social networks and it's often used to determine credit scores with alternative data***1**.

This model shows that *increasing interest rates up to a certain threshold make it more likely that the user will fall into an inescapable cycle of debt.*

Building Process: Steps 1,2 4 required a heavy iterative process.

1- Created a Bayes Net for the probability of expected Friendship and **created rules and weights for each variable and random variable** accordingly. Rules for Amount Owed Score and Payments History were also created. Page 3 shows this. This was a complex and creative task, it was also the most aligned with CS109 and my economics training.

2- In order **to have full control** and understanding of my model, I decided to **code a python version of the game**: Click here to see it in Google Colab. This was the hardest part of the project because of ensuring that the combination of parameters and weights gave the desired output.

3- Drew by hand the **main character** of the app: Chill Guy using ibisPaint X software.

4- Leveraged Claude to convert my pure Python game into a Streamlit web app with visuals. Streamlit is a platform that turns Python scripts into interactive data apps. ChatGPT was not useful for this task, after several hours of trials. Copy of my code in Google Colab here. (677 lines of code)

For a more detailed explanation of the process, click here: **YOUTUBE VIDEO**

***1** The source of inspiration for using social networks to help assess credit score was the following research paper, as well as past conversations with Stanford Development Economist Dr. Arun Chandrasekhar.
MLA Citation: Oskarsdóttir, María, et al. *Credit Scoring for Good: Enhancing Financial Inclusion with Smartphone-Based Microlending*.

Additional info and Bayesian Net: **Higher credit score → larger loan → higher expected profit, creating a positive feedback loop.**
The **expected credit score from the friends network** is calculated using a **weighted and normalized sum** by adding up the credit score of the friends times the probability of being friends.
ExpectedCreditScoreFriendships = sum(Friend_i.CreditScore * P(Friends_i)) / sum(P(Friends_i))
FinalCreditScore= 0.33 * ExpectedCreditScoreFriends + 0.33 * PaymentScore + 0.33 * (AmountOwedScore * 20) + (TotalSavings // 10)
Loan ~ N(FinalCredit * 0.4, std=15)
Profit ~ N(Loan + FinalCredit // 10, std=20)
Loan with interest = Loan * (1 + interest_rate)

